Shi Jie Samuel Tan

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Introduction

I am a PhD candidate at the Joint Center for Quantum Information and Computer Science (QuICS) at the University of Maryland, College Park. I am broadly interested in quantum error correction (QEC) and fault tolerance. I utilize concepts and tools in quantum information theory, homological algebra, and graph theory to understand and advance the construction of quantum error correcting codes and fault-tolerant protocols.

Education

University of Maryland, College Park — College Park, MD
Ph.D. in Computer Science; Advisors: Daniel Gottesman and Michael J. Gullans — 2023–2027 (Expected)

Haverford College — Haverford, PA
B.S. in Computer Science and Physics — 2019–2023
Magna Cum Laude; High Honors in Computer Science and Physics

Work

University of Maryland, College Park — College Park, MD
Research Assistant; Supervisors: Daniel Gottesman and Michael J. Gullans — 2023–Current

Google Quantum AI — Playa Vista, CA
Student Researcher; Mentors: Noah Shutty and Michael Newman — Fall 2026

IBM Quantum — Yorktown Heights, NY
QEC Theory Research Intern; Mentors: Guanyu Zhu and Theodore J. Yoder — Summer 2026

University of Maryland, College Park — College Park, MD
Grader and Teaching Assistant — 2025–2026

QuEra Computing Inc. — Boston, MA
QEC Research Intern; Mentors: Hengyun (Harry) Zhou and Arpit Dua — Summer & Fall 2025

Los Alamos National Laboratory — Los Alamos, NM
Quantum Algorithms Research Fellow; Mentors: Yiğit Subaşı and Andrew Sornborger — Summer 2023

California Institute of Technology — Pasadena, CA
QEC SURF Fellow; Mentors: John Preskill and Christopher A. Pattison — Summer 2022

University of Maryland, College Park — College Park, MD
REU-CAAR Summer Researcher; Mentors: Matthew Coudron and Gorjan Alagic — Summer 2021

Haverford College Haverford, PA
Computational Biology Undergraduate Researcher; Supervisor: Sara Mathieson 2020–2023

Haverford College Haverford, PA
Grader and Teaching Assistant 2020–2023

Publications & Preprints

♦ denotes equal contribution. ♠ denotes alphabetical order.

[1] ♠ Matthew Coudron, Michael J. Gullans, Jon Nelson, Joel Rajakumar, and **S.J.S.T.**, "Polynomial-time additive-error estimation of output probabilities for shallow quantum circuits", arXiv preprint arXiv:2610.02146 (2026).

[2] Guanyu Zhu, **S.J.S.T.**, Ryohei Kobayashi, and Po-Shen Hsin, "Sipser-Spielman meets Dijkgraaf-Witten: non-Abelian qLDPC codes via twisted sheaf gauge theory and almost-constant-overhead magic state fountain", arXiv preprint arXiv:2609.31541 (2026).

[3] Jin Ming Koh♦, Shayan Majidy♦, Aranya Chakraborty, Anqi Gong, **S.J.S.T.**, and Norman Y Yao, "Achieving the limits of automorphism gates", arXiv preprint arXiv:2609.19250 (2026).

[4] **S.J.S.T.**♦, Ian Gill♦, Eric Huang♦, Pengyu Liu, Chen Zhao, Hossein Dehghani, Aleksander Kubica, Hengyun Zhou, and Arpit Dua, "Generalized matching decoders for 2D topological translationally-invariant codes", arXiv preprint arXiv:2603.05402 (2026).

[5] Pengyu Liu, **S.J.S.T.**, Eric Huang, Umut A Acar, Hengyun Zhou, and Chen Zhao, "Low-Overhead Surface Code with Delayed Atom-Loss Detection via Pauli Envelope", arXiv preprint arXiv:2603.04156 (2026).

[6] **S.J.S.T.**♦, Yifan Hong♦, Ting-Chun Lin♦, Michael J. Gullans, and Min-Hsiu Hsieh, "Single-Shot Universality in Quantum LDPC Codes via Code-Switching", arXiv preprint arXiv:2510.08552 (2025), [**Accepted as a short plenary talk at QIP 2026**].

[7] Yifan Hong♦, **S.J.S.T.**♦, Maryam Mudassar, Noah Berthusen, and Michael J. Gullans, "Automorphism gadgets in homological product codes", arXiv preprint arXiv:2508.04794 (2025).

[8] Noah Berthusen, **S.J.S.T.**, Eric Huang, and Daniel Gottesman, "Adaptive Syndrome Extraction", PRX Quantum **6**, 030307 (2025), [**Accepted as a talk at TQC 2025**].

[9] **S.J.S.T.** and Lev Stambler, "Effective Distance of Higher Dimensional HGPs and Weight-Reduced Quantum LDPC Codes", Quantum **9**, 1897 (2025).

[10] Dhrumil Patel♦, **S.J.S.T.**♦, Yiğit Subaşı, and Andrew T. Sornborger, "Optimal Coherent Quantum Phase Estimation Via Tapering", PRX Quantum **7**, 020302 (2026).

[11] **S.J.S.T.**, Christopher A. Pattison, Matt McEwen, and John Preskill, "Resilience of the surface code to error bursts", Phys. Rev. A **113**, 022450 (2026).

[12] **S.J.S.T.**, Huyen Trang Dang, Sarah Keim, Maja Bucan, and Sara Mathieson, "Identity-by-descent (IBD) segment outlier detection in endogamous populations using pedigree cohorts", bioRxiv, 2024–08 (2024).

[13] ♠ Yongtao Deng and **S.J.S.T.**, "Random Walks on the Generalized Symmetric Group: Cutoff for the One-sided Transposition Shuffle", arXiv preprint arXiv:2211.10462 (2022).

[14] Suchetan Dontha♦, **S.J.S.T.**♦, Stephen Smith, Sangheon Choi, and Matthew Coudron, "Approximating Output Probabilities of Shallow Quantum Circuits Which Are Geometrically-Local in Any Fixed Dimension", in Proceedings of the 17th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2022) (2022), [**Accepted as a talk at TQC 2022**].

[15] Kelly Finke, Michael Kourakos, Gabriela Brown, Huyen Trang Dang, **S.J.S.T.**, Yuval B. Simons, Shweta Ramdas, Alejandro A. Schäffer, Rachel L. Kember, Maja Bućan, and Sara Mathieson, "Ancestral haplotype reconstruction in endogamous populations using identity-by-descent", PLOS Computational Biology, edited by Degui Zhi, 10.1371/journal.pcbi.1008638 (2021).

Talks

Invited talks are indicated by ★.

★ National University of Singapore (NUS) Centre for Quantum Technologies (CQT) — September 17, 2026
Generalized matching decoders for 2D topological translationally-invariant codes

★ IBM QITEC Summer Workshop — June 3, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

★ USYD-PsiQuantum Seminar — April 16, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

• APS Global Physics Summit — March 17, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

★ Harvard University (Lukin Group) — March 6, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

• 29th Annual Quantum Information Processing Conference **(Short Plenary)** — January 26, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

★ Freie Universität Berlin (Eisert Group) — November 18, 2025
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

• Joint Quantum Institute (JQI) Friday Quantum Seminar (College Park, MD) — October 3, 2025
Adaptive Syndrome Extraction

• 20th Conference on the Theory of Quantum Computation, Communication and Cryptography — September 18, 2025
Adaptive Syndrome Extraction

★ National University of Singapore (NUS) Centre for Quantum Technologies (CQT) — May 22, 2025
Automorphism gadgets in homological product codes

★ University of Maryland Communication, Control and Signal Processing (CCSP) Seminar — October 17, 2024
Effective Distance of Higher-Dimensional HGPs and Weight-Reduced Quantum LDPC Codes

• Caltech SFP Summer Seminar Day — August 18, 2022
Proving the existence of an accuracy threshold for the Union-Find decoder

• University of Maryland Research Experience for Undergraduates Combinatorics and Algorithms for Real Problems (REU-CAAR) Research Presentation — August 13, 2021
Simulation of Low-Depth Quantum Circuits

Posters

• 8th International Conference on Quantum Error Correction — June 11, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

• 8th International Conference on Quantum Error Correction — June 11, 2026
Generalized matching decoders for 2D topological translationally-invariant codes

• 8th International Conference on Quantum Error Correction — June 9, 2026
Achieving Optimal-Distance Atom-Loss Correction via Pauli Envelope

• Monte Verità Quantum Codes — April 21, 2026
Single-Shot Universality in Quantum LDPC Codes via Code-Switching

- 29th Annual Conference on Quantum Information Processing — January 26, 2026
  Automorphism gadgets in homological product codes
- 7th International Conference on Quantum Error Correction — August 12, 2025
  Automorphism gadgets in homological product codes
- 7th International Conference on Quantum Error Correction — August 11, 2025
  Adaptive Syndrome Extraction
- 28th Annual Conference on Quantum Information Processing — February 26, 2025
  Effective Distance of Higher-Dimensional HGPs and Weight-Reduced Quantum LDPC Codes
- 19th Conference on the Theory of Quantum Computation, Communication and Cryptography — September 11, 2024
  Tapered Quantum Phase Estimation
- 6th International Conference on Quantum Error Correction — November 1, 2023
  Resilience of the surface code to error bursts
- 13th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics — August 7, 2022
  Comparison of cohort-based identical-by-descent (IBD) segment finding methods for endogamous populations
- 25th Annual Conference on Quantum Information Processing — March 10, 2022
  Approximating Output Probabilities of Shallow Quantum Circuits which are Geometrically-local in any Fixed Dimension
- 7th–8th International Conference on Algorithms for Computational Biology — November 9, 2021
  Improvements to ancestral haplotype reconstruction in pedigrees
- Research in Computational Molecular Biology – 25th Annual International Conference — August 29, 2021
  Improvements to ancestral haplotype reconstruction in pedigrees
- Haverford College Marian E. Koshland Integrated Natural Sciences Center (KINSC) Undergraduate Science Research Symposium — September 10, 2020
  Validating Ancestral Haplotype Reconstruction In Endogamous Populations using Identical-by-Descent

Fellowships and Awards

- National University of Singapore Development Grant — 2024–2027
- MathQuantum Graduate Fellowship (University of Maryland) — 2023
- QuICS Lanczos Graduate Fellowship (University of Maryland) — 2023–2025
- Louis B. Green Prize in Physics and Astronomy (Haverford College) — 2023
- Phi Beta Kappa — 2023
- Los Alamos National Laboratory Quantum Computing Summer School Fellowship — 2023
- Singloh Hsu Scholarship (Haverford College) — 2023
- Computing Research Association (CRA) Outstanding Undergraduate Researcher Award Finalist — 2023
- Caltech Summer Undergraduate Research Fellowship — 2022
- Marian E. Koshland Integrated Natural Sciences Center Summer Scholarship (Haverford College) — 2022
- An Zhu-Google University of Maryland REU-CAAR Fellowship — 2021
- Brian Kovaric Fellowship (Haverford College) — 2020
- Prime Minister's Book Prize (Ministry of Education, Singapore) — 2016

Other attendance

- Los Alamos Quantum Computing Summer School — June–August 2023

Teaching (as TA or Grader)

- Computer Science 858G (University of Maryland, College Park) — Spring 2026
  Quantum Error Correction and Fault Tolerance
- Computer Science 657 (University of Maryland, College Park) — Fall 2025
  Introduction to Quantum Information Processing
- Physics H302 (Haverford College) — Spring 2023
  Advanced Quantum Mechanics
- Computer Science B340 (Bryn Mawr College) — Fall 2022
  Analysis of Algorithms
- Physics H214 (Haverford College) — Spring 2022
  Introductory Quantum Mechanics
- Computer Science H345 (Haverford College) — Spring 2022
  Theory of Computation
- Mathematics Question Center — Fall 2021–Spring 2022
  Calculus, Linear Algebra, and Abstract Algebra
- Physics H213 (Haverford College) — Fall 2021
  Waves and Optics
- Mathematics H121 (Haverford College) — Spring 2021
  Multivariable Calculus
- Computer Science H311 (Haverford College) — Fall 2020
  Computer Security: Attacks and Defenses

Reviewing

I have been invited to review articles and abstracts for the following conferences and journals.

- Annual Quantum Information Processing Conference (QIP)
- Conference on Theory of Quantum Computation, Communication and Cryptography (TQC)
- Physical Review X Quantum

Computer skills

Ordered from most to least familiar.

- **Languages**: Python, Julia, SageMath
- **QC toolkits**: Stim

References

- **Dr. Daniel Gottesman**, Brin Family Endowed Professor in Theoretical Computer Science (Professor)
  Joint Center for Quantum Information and Computer Science, University of Maryland, College Park
  dgottesm@umd.edu

- **Dr. Michael J. Gullans**, Adjunct Assistant Professor and Director of Quantum Error Correction
  Joint Center for Quantum Information and Computer Science, University of Maryland, College Park
  QuEra Computing Inc.
  mgullans@umd.edu

- **Dr. Hengyun (Harry) Zhou**, Thomas D. and Virginia W. Cabot Professor (Assistant Professor)
  Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology
  hyzhou@mit.edu

- **Dr. Guanyu Zhu**, Senior Research Scientist
  Quantum Information Theory and Error Correction Team, IBM T.J. Watson Research Center
  guanyu.zhu@ibm.com